

Intermediate Microeconomics — Week 3, Day 4

In-class Exercises: Externalities, Public Goods, the Commons

work in pairs • solutions at the end

Exercise 1 — A negative externality [competency: efficient quantity, DWL, Pigouvian tax]

A competitive chemical market has (inverse) demand and private marginal cost

$$P = 100 - Q \quad (\text{marginal benefit}), \quad MPC = 20 + Q.$$

Each unit produced emits pollution causing a constant marginal external cost $MEC = 30$.

- (a) Find the market (private) equilibrium quantity and price.
 - (b) Write the marginal social cost MSC and find the efficient quantity Q^* .
 - (c) Compute the deadweight loss of the unregulated market.
 - (d) What per-unit Pigouvian tax restores efficiency? How much revenue does it raise?
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Exercise 2 — A public good and a common resource [competency: Samuelson condition, free-riding, commons]

Part A — Public good. A shared flood barrier of size Q benefits two firms, with

$$MB_A = 50 - Q, \quad MB_B = 30 - Q, \quad MC = 40.$$

- (a) Using the Samuelson condition, find the efficient size Q^* .
- (b) If each firm contributes voluntarily (providing up to where its *own* marginal benefit equals cost), how much is provided in total? Who free-rides? Compare with Q^* .

Part B — Common fishery. B boats fish a common ground. The value per boat is $120 - 2B$ and the cost per boat is 40.

- (c) Find the number of boats a *single owner* would choose (the efficient B^*).
 - (d) Find the open-access number of boats B_{oa} . By how much is the fishery over-used, and what externality causes it?
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Reconvene: in each case state the efficient quantity, the market outcome, and the one price signal that is missing.